TABLEAU DATA ANALYSIS PROJECT

Electric Vehicle Population Data Analysis

Interactive Tableau Dashboard — Insights Report

Prepared by: Minna Nourin
Platform: Tableau Public
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150,422+	~67.83 mi	BEV + PHEV	2023
Total EVs Tracked	Avg. Electric Range	Vehicle Types Covered	Peak Adoption Year

1. Executive Summary

This report presents a comprehensive analysis of the Electric Vehicle (EV) Population dataset sourced from the Washington State Department of Licensing, visualized through an interactive Tableau dashboard. The dataset captures registered Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs) across the United States, providing a detailed snapshot of the EV landscape.

The dashboard synthesizes key performance indicators — total vehicle count, average electric range, BEV vs. PHEV split, manufacturer market share, geographic distribution, and Clean Alternative Fuel Vehicle (CAFV) eligibility — into an intuitive, filterable interface. The analysis uncovers strong growth momentum in EV adoption, dominance by a handful of manufacturers, and concentrated geographic uptake in specific states.

2. Project Objectives

- **Understand EV growth trends:** Analyze year-over-year EV registration data to identify growth trajectories and inflection points in adoption.
- **Compare BEV vs. PHEV distribution:** Quantify the proportion of fully electric versus plug-in hybrid vehicles to gauge consumer preferences.
- **Map geographic concentration:** Visualize state-wise and county-wise EV distribution to identify high-adoption regions and infrastructure needs.
- **Identify leading manufacturers:** Determine which automakers dominate the EV market and which models are most popular among consumers.
- **Assess CAFV eligibility:** Examine Clean Alternative Fuel Vehicle eligibility across the fleet to inform policy and incentive decisions.
- **Evaluate electric range performance:** Calculate average electric range to assess real-world usability and charging concerns.

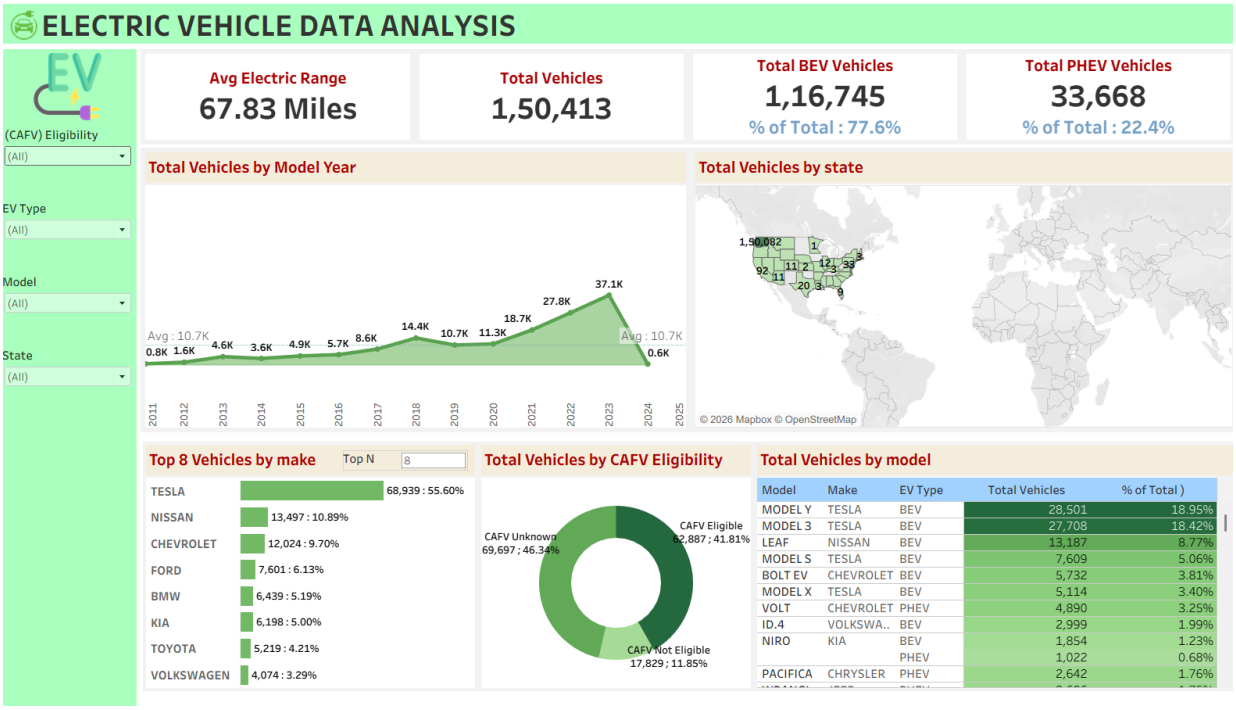
3. Dataset Overview

The dataset used for this project is the Washington State Electric Vehicle Population Data, publicly available on Data.gov. It contains registration records for EVs currently registered in the state, with each row representing a single vehicle.

Attribute	Description
VIN	Vehicle Identification Number (partial)
County / City	Geographic location of registration
State	US state of registration
Model Year	Manufacturing year of the vehicle
Make / Model	Manufacturer and model name
Electric Vehicle Type	BEV (Battery) or PHEV (Plug-in Hybrid)
CAFV Eligibility	Clean Alternative Fuel Vehicle eligibility status
Electric Range	Maximum range on electric power (miles)
Legislative District	WA legislative district of registration

4.Dashboard Components & Visualizations

The Tableau dashboard is organized into multiple interconnected visualization panels, each targeting a specific analytical dimension of the EV dataset. All charts support cross-filtering and drill-down interactions.



3.1 KPI Scorecards

- Four headline KPI cards are displayed at the top of the dashboard.
- Metrics include: Total Vehicles, Average Electric Range, Total BEV Vehicles, and Total PHEVVehicles.
- These provide instant orientation for stakeholders before exploring deeper visuals.

3.2 EV Growth by Model Year (Line/Area Chart)

- An area chart plots the count of registered EVs against model year.
- The chart reveals exponential growth starting around 2016, with a sharp peak in 2023.
- This visualization highlights the acceleration of EV adoption over the past decade.

3.3 Geographic Distribution (Map)

- A choropleth or filled map displays EV registration density by US state.
- Washington State shows the highest concentration, reflecting the dataset's origin.
- The map supports zoom, hover tooltips, and state-level filtering.

3.4 Top 10 EV Makes (Bar Chart)

- A horizontal bar chart ranks the top 10 vehicle manufacturers by registration count.
- Tesla dominates significantly, followed by Nissan, Chevrolet, Ford, and BMW.
- This chart is sortable and linked to the model-level detail view.

3.5 EV Type Distribution (Donut/Pie Chart)

- A donut chart illustrates the BEV vs. PHEV split across the total fleet.
- BEVs account for a substantial majority, reflecting growing consumer confidence in fully electrictechnology.

3.6 CAFV Eligibility Breakdown

- A bar or pie chart segments the fleet by CAFV eligibility status: Eligible, Not Eligible, and Unknown.
- A large portion falls in the 'Unknown' or 'Not Eligible' category, pointing to gaps in incentiveutilization.

3.7 Top 10 Models (Tree Map or Bar Chart)

- Displays the most popular EV models within the dataset.
- Tesla Model Y and Model 3 rank at the top, underlining Tesla's market dominance.
- Other notable entries include Nissan Leaf, Chevrolet Bolt EV, and Ford Mustang Mach-E.

5.Key Findings & Insights

Finding 1	**Tesla's Market Dominance**
	Tesla accounts for the largest share of registered EVs, with the Model Y and Model 3 being the two most popular models. This dominance reflects Tesla's early-mover advantage, broad Supercharger network, and strong brand recognition among EV adopters.
Finding 2	**Strong Growth Trajectory**
	EV registrations show consistent year-over-year growth from 2011 to 2023, with the sharpest increase in recent years. The peak in 2023 suggests that mainstream EV adoption is well underway, driven by falling battery costs and expanding model availability.
Finding 3	**BEV Majority**
	Battery Electric Vehicles (BEVs) constitute the majority of the registered fleet, surpassing Plug-in Hybrid Electric Vehicles (PHEVs). This trend reflects consumer preference for zero-emission vehicles over hybrid alternatives as range anxiety diminishes.
Finding 4	**Geographic Concentration in Washington State**

	The dataset is heavily weighted towards Washington State registrations. Within Washington, certain counties — particularly King County (Seattle metropolitan area) — show disproportionately high EV ownership, correlating with higher income levels and better charging infrastructure.
Finding 5	**Average Range of ~67.83 Miles**
	The average electric range across all vehicles in the dataset is approximately 67.83 miles. While newer BEV models offer significantly higher ranges (250-350+ miles), older PHEVs and early BEVs pull this average down, indicating that range improvement over time is a key trend.
Finding 6	**CAFV Eligibility Gap**
	A significant portion of the fleet is either CAFV-ineligible or has unknown eligibility status. This suggests opportunities for policy expansion and better tracking of clean fuel incentives to encourage further EV adoption.

6.Tools & Methodology

Tool / Technology	Purpose
Tableau Desktop / Public	Primary visualization and dashboard creation platform
Washington State DOL Dataset	Source data (CSV) containing EV registration records
Calculated Fields (Tableau)	Derived KPIs: Total BEV, Total PHEV, Avg. Electric Range
Sets & Filters	Top-N filtering (e.g., Top 10 Makes, Top 10 Models)
Dashboard Actions	Cross-filter interactivity between all charts
GitHub	Version control and public portfolio hosting

7.Recommendations

■ Expand Charging Infrastructure in Low-Adoption States

States with low EV registration counts could benefit significantly from targeted charging infrastructure investment. Policy makers should use geographic distribution maps to prioritize high-impact corridors.

■ Incentivize CAFV-Eligible Vehicle Purchases

Increasing awareness and streamlining the CAFV eligibility verification process could encourage more consumers to take advantage of clean fuel incentives, further accelerating adoption.

■ Support Diverse Manufacturers

While Tesla dominates current registrations, supporting a broader ecosystem of EV manufacturers through targeted incentives can foster competition, reduce prices, and increase choice for consumers.

■ Address Range Anxiety Through Education

The average electric range statistic, combined with model-year trends showing steady improvement, can be used in consumer education campaigns to address lingering range anxiety concerns.

8.Conclusion

This project demonstrates the power of Tableau as a data storytelling tool applied to a real-world, policy-relevant dataset. The Electric Vehicle Population dashboard transforms raw registration records into actionable intelligence — revealing market trends, geographic patterns, and manufacturer dynamics that are critical for stakeholders ranging from policymakers to automotive investors.

The consistent growth in EV registrations, the clear dominance of BEV technology over PHEVs, and the geographic concentration of adoption in urban, high-income regions all point toward a rapidly evolving transportation landscape. As battery technology improves and charging infrastructure expands, these trends are expected to deepen and broaden geographically.

This dashboard serves as both a portfolio piece demonstrating proficiency in Tableau and as a meaningful analytical contribution to the discourse on sustainable transportation.